

Business Requirements Document

Aegis Resolve AI — Digital Arbitration Court

Field	Value
Product	Aegis Resolve AI
Version	MVP 0.1.0
Document Owner	Business & Product Team
Date	July 27, 2026
Classification	Internal — Demo Submission
Status	Approved

1. Executive Summary

Aegis Resolve AI addresses the critical business need for faster, fairer, and more transparent payment dispute resolution. The platform models card and payment disputes as a Digital Arbitration Court, providing evidence-led decisions with full auditability. This MVP demonstrates the business viability of AI-assisted dispute resolution to stakeholders including card issuers, merchants, and compliance teams.

2. Business Problem

Payment disputes represent a significant operational cost and customer satisfaction risk for financial institutions. Current processes are characterized by: (a) resolution timelines of 2–6 weeks, (b) opaque decision-making that erodes customer trust, (c) inconsistent analyst quality under high volume, (d) incomplete audit trails for regulatory compliance, and (e) adversarial merchant-cardholder dynamics without structured evidence evaluation.

3. Business Objectives

Primary objectives for the Aegis Resolve platform include: reduce average dispute resolution time from weeks to minutes; increase decision transparency with policy citations and explainable rationale; maintain fairness parity above 90% across all case types; provide 100% auditable replay trails for compliance review; and route ambiguous edge cases to human analysts with full context preserved.

4. Stakeholders

Stakeholder	Role	Interest
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Card Members	Consumers	Fast, fair resolution of disputed charges
Merchants	Businesses	Evidence-based defense, reduced chargeback losses
Dispute Analysts	Operations	Efficient queue management, override capability
Compliance Officers	Regulatory	Complete audit trails, policy adherence
Product Leadership	Strategy	Platform differentiation, cost reduction
Engineering	Delivery	Scalable, maintainable architecture

5. Business Requirements

ID	Requirement	Priority
BR-01	Support card member dispute filing and status tracking	Must Have
BR-02	Enable merchant evidence submission and case response	Must Have
BR-03	Provide enterprise analyst review queue with override	Must Have
BR-04	Generate policy-grounded verdicts with AMEX reason codes	Must Have
BR-05	Score evidence trust and case fairness (0–100)	Must Have
BR-06	Maintain complete courtroom replay audit trail	Must Have
BR-07	Route ambiguous cases to human review automatically	Must Have
BR-08	Deliver operational KPI dashboard for analysts	Should Have
BR-09	Support real-time event streaming for live replay	Should Have
BR-10	Integrate with payment network evidence sources	Future
BR-11	Deploy LLM agent orchestration for reasoning	Future
BR-12	Implement authentication and tenant isolation	Future

6. Success Metrics & KPIs

Metric	Current Baseline	MVP Target	Production Target
Avg Resolution Time	2–6 weeks	≤ 8 minutes	≤ 15 minutes
Fairness Index	Unmeasured	≥ 90 avg	≥ 95 avg
Audit Completeness	Partial	100%	100% immutable
Human Review Rate	100% manual	40% (edge cases)	≤ 15%
Customer Satisfaction	Low (opaque)	Demo validated	≥ 4.5/5
Compliance Pass Rate	Variable	Full replay	Automated checks

7. Business Constraints

- MVP must run locally without external API keys or cloud dependencies.
- Demo must include five pre-seeded scenarios covering all AMEX reason codes.
- No production authentication required for hackathon/demo submission.
- AMEX reason codes used as policy proxy — not affiliated with American Express.
- Agent pipeline is conceptual in MVP; real LLM integration deferred to production.
- Budget constraint: zero external service costs for demo phase.

8. Assumptions & Dependencies

- Stakeholders have access to modern browsers for demo presentation.
- Python 3.12 and Node.js are available on demo machines.
- Compliance team will validate policy code mappings before production.
- LLM provider selection requires enterprise security review.
- Payment network integration requires partnership agreements.

9. Risks & Mitigations

Risk	Impact	Mitigation
LLM hallucination in verdicts	High	Policy engine guardrails + human review routing
Regulatory non-compliance	High	Immutable audit store + policy versioning

Merchant adoption resistance	Medium	Transparent evidence graph + dual advocacy
Demo reliability issues	Medium	Seeded data + frontend fallback + Docker
Scalability bottlenecks	Low (MVP)	PostgreSQL + microservices roadmap defined

10. Approval

Role	Name	Signature	Date
Product Owner	Dictatordaddy	Approved	July 27, 2026
Technical Lead	Dictatordaddy	Approved	July 27, 2026
Business Sponsor	Pending		

— End of Business Requirements Document —